TRIALGPT LITE V2.0 | 100% OFFLINE ENGINE | EXECUTION < 1MS | GRADE 6-8 READING LEVEL

Clinical Trial Eligibility Screener

Architecture Diagrams, Decision Trees & Evaluation Technical Specification

Component: `trialgpt_lite.parser.question_generator` **Service:** FastAPI / ClinicalTrials.gov REST API v2 **Artifact Path:** `docs/QUESTIONNAIRE_EVALUATION_ARCHITECTURE.md`

1. High-Level System Architecture

The **Interactive Eligibility Screener** translates complex clinical trial eligibility criteria into plain, easy English questions (Grade 6–8 health literacy level) and evaluates candidate answers 100% offline in sub-milliseconds without external LLM dependencies.

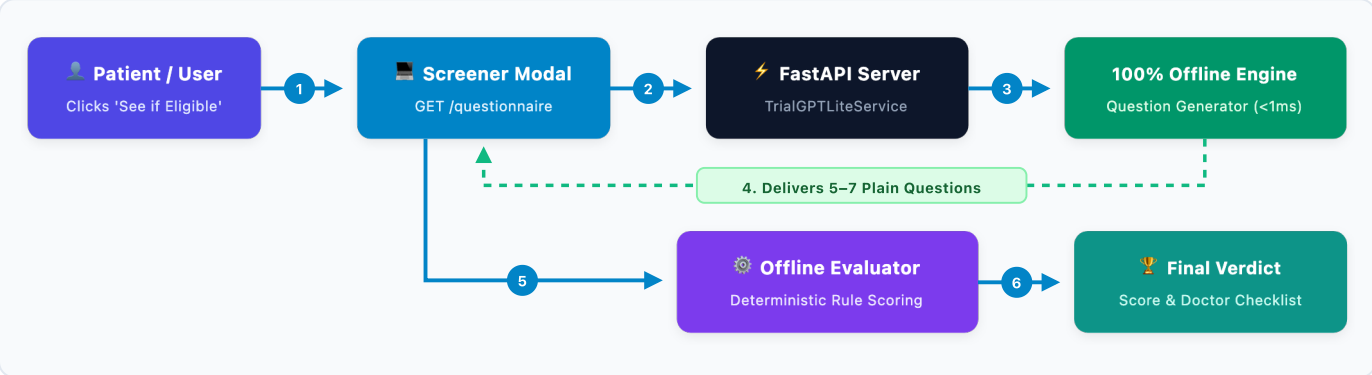


Figure 1.1: End-to-End System Flowchart — User interaction through offline rule generation and instant verdict delivery.

2. System Sequence Diagram (Phases 1 & 2)

The system executes in two decoupled, stateless phases: Questionnaire Generation (Phase 1) and Instant Rule Evaluation (Phase 2):

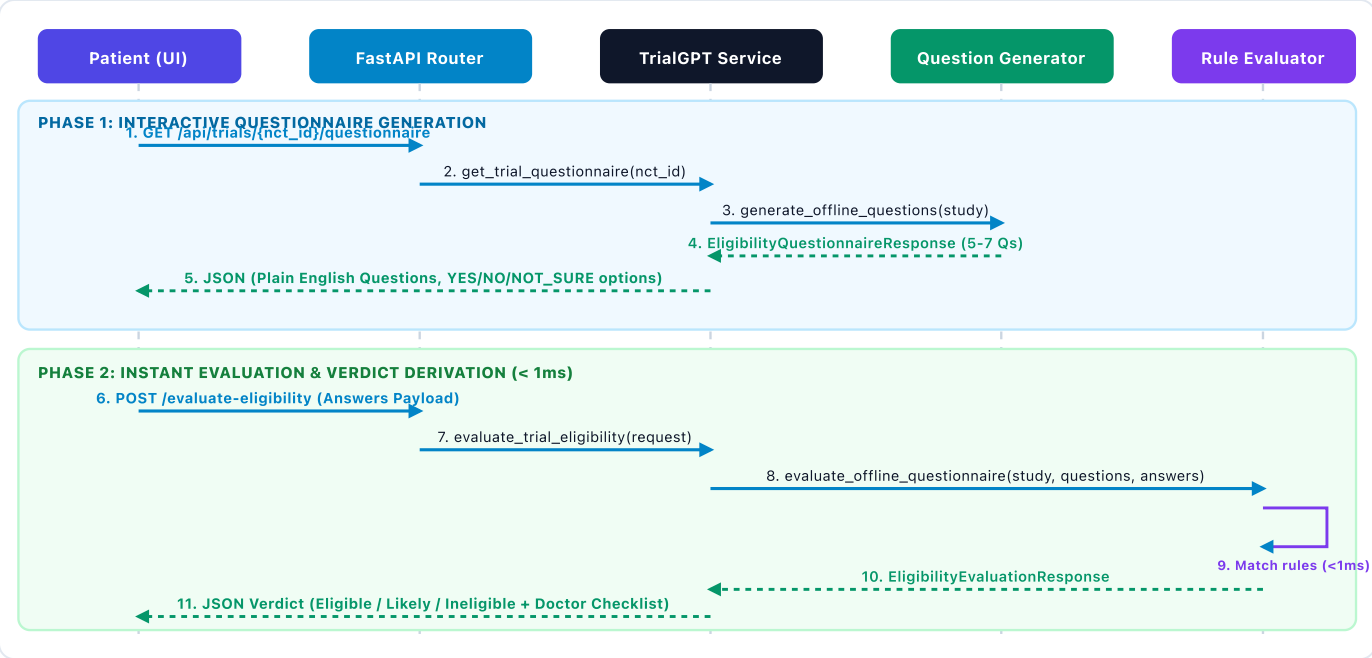


Figure 2.1: Complete Execution Sequence Diagram — Phase 1 (Criteria Parsing) and Phase 2 (Deterministic Evaluation).

3. Criteria Classification & 3-State Scoring Rules

Every protocol requirement is categorized into **Inclusions** (required conditions) or **Exclusions** (disqualifying conditions).

RULE CATEGORY	CLINICAL DEFINITION	TARGET ANSWER	CLINICAL IMPACT
Inclusion Criteria	Conditions the patient *must have* (e.g., confirmed cancer diagnosis, advanced/metastatic stage, prior treatment).	**YES**	Answering "NO" triggers an instant protocol disqualification.
Exclusion Criteria	Conditions the patient *must not have* (e.g., active brain tumors, severe heart failure, active viral infection, pregnancy).	**NO**	Answering "YES" triggers an instant study exclusion.

The 3 Evaluation States

USER ANSWER	CRITERIA TYPE	STATE	VISUAL STATUS	CLINICAL ASSESSMENT
YES	INCLUSION	`MET`	MET ✓	Patient meets mandatory requirement.
NO	EXCLUSION	`MET`	MET ✓	Patient does not have prohibited condition.
NOT_SURE	ANY	`UNCERTAIN`	UNCERTAIN ?	Patient is unsure. Routed to Doctor Consultation Checklist.
NO	INCLUSION	`UNMET`	UNMET ✕	Lacks mandatory condition (Disqualification).
YES	EXCLUSION	`UNMET`	UNMET ✕	Has prohibited condition (Disqualification).

4. Hierarchical Decision Tree & Mathematical Scoring

The evaluator tallies all evaluated states and resolves the final outcome via a strict hierarchical decision tree:

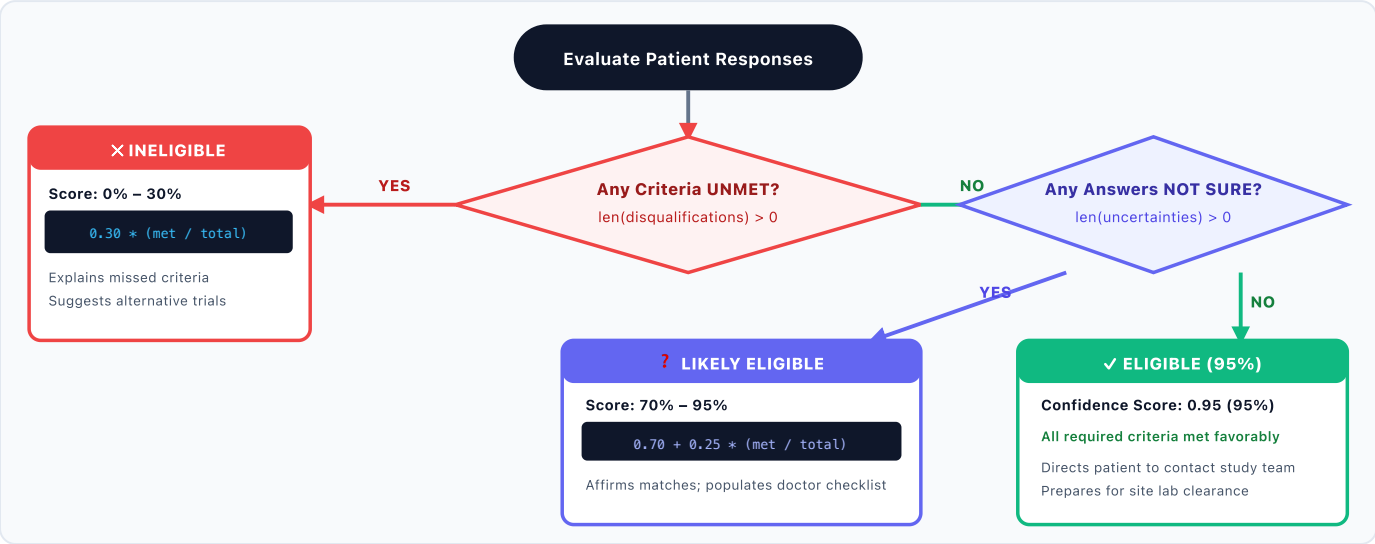


Figure 4.1: Hierarchical Decision Tree Diagram — Resolution logic for INELIGIBLE, LIKELY_ELIGIBLE, and ELIGIBLE outcomes.

5. Dynamic Doctor Consultation Checklist Derivation

Whenever a patient selects **"Not Sure / Ask Doctor"**, the engine extracts the core clinical concept and synthesizes a tailored consultation agenda combined with baseline trial protocols:

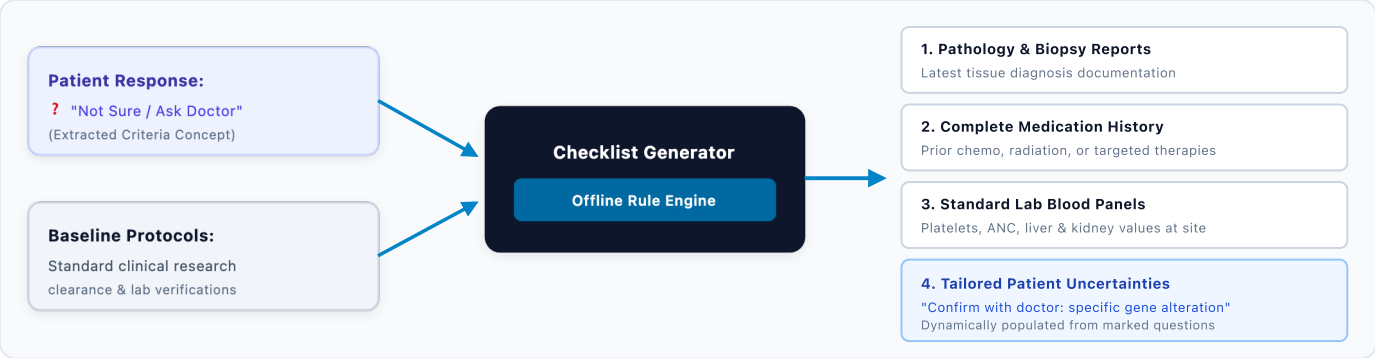


Figure 5.1: Doctor Checklist Synthesis Flow — Combining baseline protocol verifications with patient-specific uncertainties.

5.1 Three-Pillar Synthesis Strategy

PILLAR	PURPOSE	SAMPLE OUTPUT ITEM
1. Diagnostic Verification	Ensure pathology, tumor histology, and stage match protocol requirements.	*"Bring your latest pathology and biopsy reports to your consultation."*
2. Treatment History	Verify prior systemic chemotherapy, radiation, or immunotherapy line numbers.	*"Provide complete medication history and prior chemotherapy records."*
3. Laboratory Clearance	Assess organ function (liver enzymes, creatinine, platelet/ANC counts).	*"Standard lab blood counts (platelets, liver panels) verified at site."*
4. Patient Uncertainties	Direct conversion of items marked `NOT_SURE` into doctor questions.	*"Confirm with doctor: whether prior chemotherapy history meets study interval."*

> **Patient Experience Philosophy:** The screener is designed to empower, never discourage. Even when marked `LIKELY_ELIGIBLE` or with uncertainty, patients leave with a tangible, physician-ready consultation sheet so their clinical care team can immediately review clinical study enrollment without administrative friction.

6. Key Architectural Guarantees & Production SLAs

CAPABILITY	ENGINE BEHAVIOR	CLINICAL & OPERATIONAL BENEFIT
100% Offline Generation	Rule-based clinical dictionary matching with zero external LLM calls.	Zero cost per user evaluation; no third-party downtime risk; fully compliant.
Sub-Millisecond SLA	In-memory string operations and dictionary evaluations in <1ms.	Instant UI modal response; eliminates patient drop-off during screener.
Deterministic Logic	Strict mathematical decision tree with reproducible scoring (0% to 95%).	100% explainable verdicts; prevents AI hallucination in patient screening.
Plain English Questions	Jargon normalized to Grade 6–8 reading level with simple YES/NO choices.	Enables elderly and non-technical patients to understand trial parameters.

7. Python Core Implementation Reference (question_generator.py:310-392)

Below is the complete, production-tested offline evaluation logic executing with zero network calls and deterministic behavior:

```
def evaluate_offline_questionnaire(
    study: TrialStudy,
    questions: list[EligibilityQuestion],
    answers_map: dict[str, str],
) -> EligibilityEvaluationResponse:
    question_dict = {q.id: q for q in questions}
    evaluations, disqualifications, uncertainties = [], [], []
    satisfied_count = 0

    for q_id, q in question_dict.items():
        user_answer = (answers_map.get(q_id) or "NOT_SURE").strip().upper()
        favorable = q.favorable_answer

        if user_answer == "NOT_SURE":
            status = "UNCERTAIN"
            reason = "You marked 'Not Sure'. Your doctor will verify this from your medical records."
            uncertainties.append(q.question_text)
        elif user_answer == favorable:
            status = "MET"
            reason = f"Your answer ('{user_answer.title()}') matches the criteria required for this trial."
            satisfied_count += 1
        else:
            status = "UNMET"
            if q.category == "EXCLUSION":
                reason = "You answered 'Yes', which is an exclusion criterion for this trial."
                disqualifications.append(q.question_text.rstrip("?").replace("Do you have ", ""))
            else:
                reason = "You answered 'No', but this trial specifically requires this condition."
                disqualifications.append(q.question_text.rstrip("?").replace("Have you ", ""))

        evaluations.append(CriterionEvaluationItem(
            criterion=q.question_text,
            category=q.category,
            status=status,
            user_answer=user_answer,
            reason=reason,
        ))

    total_q = len(questions) or 1
    if disqualifications:
        verdict = OverallEligibility.INELIGIBLE
        score = max(0.0, round(0.3 * (satisfied_count / total_q), 2))
        summary = (
            f"Based on your answers, you are likely ineligible for this trial because some protocol requirements "
            f"were not met ({'; '.join(disqualifications[:2])}). We encourage you to discuss alternative clinical trials "
            f"and treatment options with your oncologist or care team."
        )
    elif uncertainties:
        verdict = OverallEligibility.LIKELY_ELIGIBLE
        score = round(0.70 + 0.25 * (satisfied_count / total_q), 2)
        summary = (
            f"Good news! You meet {satisfied_count} key trial criteria. Because {len(uncertainties)} questions "
            "require confirmation from your medical records or doctor, we recommend sharing this study with your care team."
        )
    else:
        verdict = OverallEligibility.ELIGIBLE
        score = 0.95
        summary = "Great news! Based on your self-reported responses, you meet all major eligibility criteria for this trial."

    doctor_checklist = [
        "Bring your latest pathology and biopsy reports to your consultation.",
        "Provide your complete medication history and prior chemotherapy/immunotherapy records.",
        "Standard laboratory blood counts (platelets, neutrophils, liver & kidney panels) will be verified by the clinical research
    ]
    for unc in uncertainties:
        clean_unc = unc.rstrip("?").replace("Do you have ", "").replace("Have you ", "").replace("Are you ", "")
        doctor_checklist.append(f"Confirm with doctor: {clean_unc}")

    return EligibilityEvaluationResponse(
        nct_id=study.nct_id,
        verdict=verdict,
        score=score,
        summary=summary,
        criteria_evaluations=evaluations,
        doctor_checklist=doctor_checklist,
    )
```